Predicting Early-Stage Diabetes from Symptoms: A Toy Workflow Example

Nur Ceren Kabakci

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Abstract

This paper shows a reproducible workflow. A logistic regression model predicts diabetes from symptoms with 92.3% accuracy on unseen patients.

1 Introduction

Early detection od diabetes can improve patient outcomes. We use the symptom survey dataset of [1], which has 520 patients, of whom 61.5% have diabetes.

2 Methods

2.1 Data

Each patient has an age, a gender and 14 yes/no symptoms. Details of the data are given in Section S1 of the SI.

2.2 Model

I fit a logistic regression model, implemented in scikit-learn [2]. It is described by

$$P(\text{diabetes} = 1 \mid x) = \frac{1}{1e^{-(+b_0+b_1x_1+\ldots b_px_p)}}, \tag{1}$$

where $x_1, \ldots, x_p$ are the predictors and $b_0, \ldots, b_p$ are coefficients learned from the data. Equation 1 is the model used in Section 3.

3 Results

I trained on 80% of patients and tested on the remaining 20%. The test accuracy was 92.3%. Figure 1 shows the fitted coefficients; larger positive values mean that symptom raises the predicted chance of diabetes.

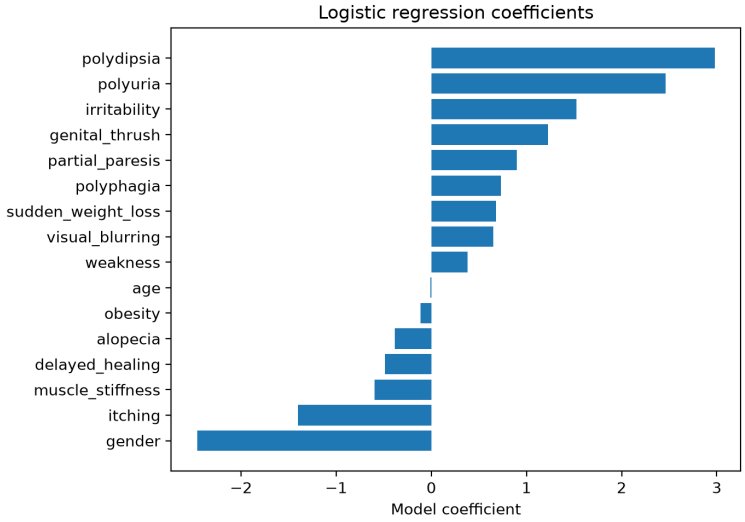


Figure 1: Coefficients of the logistic regression model. Positive bars increase the predicted probability of diabetes.

4 Discusion

This was a simple analysis and results may change with other models. More detail on the data is in Section S1 of SI.

References

[1] M. M. Faniqul Islam, Rahatara Ferdousi, Sadikur Rahman, and Humayra Yasmin Bushra. Likelihood prediction of diabetes at early stage using data mining techniques. In *Computer Vision and Machine Intelligence in Medical Image Analysis*, pages 113–125. Springer, 2020.

[2] Fabian Pedregosa et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.